

Project Structure Overview

A Mobile Application for Facility Management

Team

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Members

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Tutorial Details

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The CampusCare project is organized into two main folders:

1. backend/

The backend folder contains the Node.js and Express.js server. It is responsible for authentication, user roles, issue management, worker management, admin management, notifications, image uploads, and communication with the Supabase PostgreSQL database.

database/

The database folder contains the SQL files used to create and prepare the database.

- `schema.sql`: Contains the database structure, including tables, relationships, constraints, and required database definitions.
- `seed.sql`: Contains initial sample data or required startup data, such as categories, locations, users, and other default records.

node_modules/

The `node_modules` folder contains the installed backend dependencies. It is automatically generated after running `npm install` and should not be edited manually.

src/

The `src` folder contains the main backend source code.

config/

The `config` folder contains backend configuration files.

- `cloudinary.js`: Contains the Cloudinary configuration used to store uploaded issue photos and completion photos.

controllers/

The `controllers` folder contains the main backend logic for each feature. Controllers receive requests, process the required data, communicate with the database, and return responses to the frontend.

It includes:

- `authController.js`: Handles registration, login, password reset, and authentication-related logic.
- `issueController.js`: Handles issue submission, issue updates, assignment, status changes, comments, notifications, and issue management.
- `UserController.js`: Handles user-related functions, such as viewing users and activating or deactivating accounts.

db/

The db folder contains the database connection file.

- index.js: Connects the backend to the Supabase PostgreSQL database using the database connection string.

middleware/

The middleware folder contains functions that run before requests reach the controllers. These functions help protect routes and handle uploaded files.

It includes:

- authMiddleware.js: Verifies JWT tokens and protects private routes.
- uploadMiddleware.js: Handles image uploads before sending them to Cloudinary.

routes/

The routes folder contains the API route files. Each route file groups related backend endpoints.

It includes:

- authRoutes.js: Contains authentication routes such as register, login, logout, and password reset.
- issueRoutes.js: Contains issue-related routes such as submitting issues, viewing issues, assigning workers, updating status, adding comments, uploading completion photos, and managing notifications.
- userRoutes.js: Contains user management routes such as viewing users and activating or deactivating accounts.

.env

The .env file contains private environment variables, such as the Supabase database connection string, JWT secret, and Cloudinary credentials. This file should not be uploaded publicly because it contains sensitive information.

index.js

The index.js file is the main backend entry point. It starts the Express server, connects middleware, registers the backend routes, and runs the API.

package.json

The package.json file contains backend dependencies, scripts, and project metadata.

package-lock.json

The package-lock.json file stores the exact installed versions of backend dependencies to ensure consistent installation across different devices.

2. frontend/

The frontend folder contains the React Native Expo mobile application. It is responsible for the user interface, navigation, role-based screens, forms, image selection, and communication with the backend API.

.expo/

The .expo folder contains Expo-generated project files. It is created automatically when running the Expo application.

assets/

The assets folder contains application images, icons, and other static resources used in the mobile app.

It includes:

- icon.png: The application icon used by Expo.

node_modules/

The node_modules folder contains the installed frontend dependencies. It is automatically generated after running npm install and should not be edited manually.

src/

The src folder contains the main frontend source code.

navigation/

The navigation folder contains the files used to move between screens in the application. It also supports role-based navigation, so each user role is directed to the correct dashboard after login.

It includes:

- AdminNavigator.js: Handles navigation for System Admin screens.
- CMNavigator.js: Handles navigation for Community Member screens.
- FMNavigator.js: Handles navigation for Facility Manager screens.
- WorkerNavigator.js: Handles navigation for Worker screens.

screens/

The screens folder contains all application screens for the different user roles.

admin/

The admin folder contains System Admin screens.

- AdminUserListScreen.js: Allows the admin to view registered users and activate or deactivate user accounts.

manager/

The manager folder contains Facility Manager screens.

- FMDashboardScreen.js: Displays submitted issues and allows the manager to search, filter, and manage issues.
- FMIssueDetailScreen.js: Displays full issue details and allows the manager to assign workers, update issue status, set priority, delete issues, and manage issue progress.

member/

The member folder contains Community Member screens.

- common.js: Contains shared styling or helper elements used by member screens.
- HomeScreen.js: Displays the Community Member home screen.
- IssueDetailScreen.js: Displays full details of a selected submitted issue.
- IssueStatusScreen.js: Displays issue status information.
- MyIssuesScreen.js: Displays issues submitted by the logged-in Community Member.
- SubmitIssueScreen.js: Allows the Community Member to submit an issue with a title, description, category, location, and photo.

worker/

The worker folder contains Worker screens.

- AssignedIssuesScreen.js: Displays issues assigned to the logged-in worker.
- IssueDetailScreen.js: Displays assigned issue details and allows the worker to update progress, add comments, upload a completion photo, and resolve the issue.

Authentication Screens

These screens are used by multiple roles:

- LoginScreen.js: Allows users to log in.
- RegisterScreen.js: Allows users to create an account.
- ResetPasswordScreen.js: Allows users to reset their password using the simplified password reset process.

services/

The services folder contains files responsible for connecting the frontend to the backend API.

- **api.js**: Contains the base API URL and request functions used by the frontend to communicate with the backend.

App.js

The App.js file is the main entry point of the React Native Expo application. It connects the navigation structure and starts the mobile app.

app.json

The app.json file contains Expo configuration, such as the app name, icon, splash screen, and platform settings.

index.js

The index.js file registers and starts the Expo React Native application.

package.json

The package.json file contains frontend dependencies, scripts, and project metadata.

package-lock.json

The package-lock.json file stores the exact installed versions of frontend dependencies to keep installations consistent.